How Many Independent Models Does an Open-Model Ecosystem Contain?

One-page research brief • Shaurya Gupta • github.com/SilverCoin256/cofail

The question. If many independently developed AI models are deployed as an ensemble, a judging panel, or competing screening vendors, how much of that plurality is real? The standard evidence — how often pairs of models fail on the same test item, versus a baseline assuming independence — cannot answer this, for a provable reason.

The result. The mean co-failure rate across a model population is a function of *item difficulty alone*; it is invariant under any randomization that preserves that difficulty, so its excess over such a baseline is exactly zero, not merely small. That degeneracy turns out to be a known result independently in three fields (ecology's V-ratio, psychometrics' Cronbach's α, Fleiss's rater agreement) that had never been connected to AI evaluation. Conditioning on the one *exact, fit-free* null available — the distribution implied by the Rasch model's own sufficient statistics, requiring no estimated parameters — removes that artifact and leaves a real second-order signal: residual correlation **5.5**× the null on the primary benchmark (**2.9–11.9**× across all five), concentrated in **roughly six** extra latent dimensions, across **1,228–1,373 open models per benchmark** on five benchmarks, harvested at $0 compute cost from public evaluation logs. The exact-null sampler itself is validated by direct enumeration (a χ^2 test against uniformity on fibres up to size 1,170 does not reject), and its detection power against planted alternatives is characterised rather than assumed — including one alternative it has *no* power against by construction, stated as a property of the design rather than a caveat.

What makes this a research process rather than a result.

- A pre-registered primary hypothesis came back with an inconsistent sign across benchmarks. Its kill condition, written in advance, called that a refutation — reported as refuted, not reinterpreted.
- An adversarial review, run deliberately against the project's own headline finding before submission, found that "∼1,300 models behave like ∼24 independent ones" is algebraically incapable of distinguishing one weak shared factor from many tight clusters. **The claim was withdrawn** and replaced with the diagnostics that can tell the two apart.
- A closely related paper turned up in a deliberate prior-art search on the same archive. Rather than abandon the direction, the project narrowed to the specific open question that paper leaves unanswered.
- A separate pre-registered test, aimed at a published claim in prior work, came back against the project's own hypothesis: the claim survives conditioning almost fully (88% of the effect remains) and is reported as robust, exactly as committed to in advance.

What keeps happening after this document. A scheduled process re-checks the public archive monthly, on GitHub's own infrastructure, and appends a dated row to a public time series — the same frozen method applied to more data, not a new claim.

Full paper, pre-registration, code, and every results artifact: github.com/SilverCoin256/cofail. Claude (Anthropic) was used as a coding and analysis assistant throughout; research direction, pre-registered commitments, and final synthesis are the author's own.